**BRAC UNIVERSITY**
**Department of Computer Science and Engineering**

Examination: Quiz 3 Duration: 20 min
Semester: Summer 25 Full Marks: 15

**CSE 470: Software Engineering**

**Name:** **ID:** **Section:**

**Set-***

A real-time chat application allows users to send and receive messages instantly. The application supports different chat rooms where users can join discussions on various topics. Each chat room is constantly updating the chat window as new messages are sent by participants, ensuring that everyone in the room can see the latest conversation in real-time.The system ensures that there is one centralized messaging service that handles all the chat messages across different chat rooms. This service manages message delivery, ensures consistency, and prevents multiple instances of messaging systems running concurrently. Additionally, when a new message is sent to any of the rooms, all users currently connected to the chat room are immediately notified of the new message, and their chat windows are updated.

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|---|---|---|---|
| **1** | **CO1** | **a. Which design pattern/patterns are used here ?Why [5]**<br>**b. Write down the code of the scenario based on your previous answer ? [10]** | **15** |

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In an office, there are several different types of printers (laser, inkjet, and thermal), each with its own interface for managing print jobs. The company needs a unified system that can interact with all the printers, allowing users to send print jobs from different devices in a consistent way, regardless of the printer's type or model. The system has a centralized printer manager that is responsible for managing all print jobs. There is only one instance of the manager, which coordinates the printing tasks and maintains the settings for all printers. Users can interact with the manager, and it will take care of delegating the print jobs to the respective printers. However, since the printers use different protocols for communicating with the system, the system uses a mechanism that translates the print jobs into a format that is compatible with each specific printer's communication protocol. This ensures that the centralized printer manager can handle print jobs for all printers in a uniform manner, regardless of the type of printer.

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|---|---|---|---|
| **1** | **CO1** | **c. Which design pattern/patterns are used here ?Why [5]**<br>**d. Write down the code of the scenario based on your previous answer ? [10]** | **15** |